11. Domain: Agri Electronics

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	Voltammetric technique based electrochemical system for quality analysis of liquid samples
	Name of the Project under which technology is developed	Quality Measurement of Black Tea by Electronic Means – Integration of E-Nose, E-Tongue and E-Vision
	Funding Agency	NTFR
	Technology Readiness Level (TRL)	TRL-8
	Brief Write-up on the Technology Developed	Voltammetric method based electronic tongue has been developed for measurement of strength of black tea liquor, which is one of the most important quality parameters of black tea. Different sensors were used sequentially as working electrode where each sensor has a distinct recognition mechanism, with a different dynamic response, which contributes to the overall discrimination power of the E-tongue system
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Subrata Sarkar, C-DAC Kolkata Mobile : +91 9433111038, subrata.sarkar@cdac.in
2.	**Brief details of Technology Developed**	The present device pertains to advancements in the technical domain of device development, specifically focusing on processes for detecting mastitis in cow milk. More precisely, this invention introduces a hand-held mastitis detection device, designed to function as a field-level screening tool for identifying mastitis infections in cow milk. This innovation holds significant potential benefits for the milk production sector, cattle yards, and the broader dairy industry. Mastitis remains a prevalent and costly issue in the dairy industry, characterized by an inflammatory response in the udder tissue due to microbial infections. This disease is a leading cause of economic losses, primarily due to decreased milk yield and compromised milk quality. The development of a portable, rapid screening device for mastitis detection in cow milk addresses this critical need, offering substantial advantages to dairy producers and consumers worldwide.
	Name of the Project under which technology is developed	National Programme on Electronics and ICT Applications in Agriculture and Environment (AgriEnIcs)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-8
	Brief Write-up on the Technology Developed	The developed technology is a portable, microcontroller-based diagnostic system engineered for the rapid, in-situ detection of mastitis in dairy cattle. Utilizing a reagent-free sensing mechanism, the device eliminates the need for chemical additives, relying instead on a custom PCB integrated with Operational Amplifiers (OpAmps) for precise signal conditioning. To ensure analytical accuracy across diverse environmental conditions, specifically Indian

Sr. No.	Parameters	Brief details
		climates, the system incorporates an embedded temperature sensor for real-time thermal compensation. The device provides immediate qualitative classification of milk health ("Healthy" or "Mastitis") via a high-contrast Digital O-LED display, optimized for operation in low-visibility settings. Extensively validated over three years at ICAR-NDRI, this robust, Made in India solution requires no installation or maintenance overhead. It offers a user-friendly, high-fidelity tool for dairy monitoring, requiring only routine sensor cleaning between measurements to ensure consistent performance.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr Subhankar Mukherjee, C-DAC Kolkata 9477355692, subhankar.mukherjee@cdac.in
3.	**Brief details of Technology Developed**	RIGE-SENSE, a Machine Vision Solution for Estimation of Milling Age of Raw Rice
	Name of the Project under which technology is developed	AI-based Agri-photonics Solution for Estimation of the Age of Rice
	Funding Agency	Food Corporation of India (FCI), Ministry of Consumer Affairs, Food and Public Distribution, Government of India
	Technology Readiness Level (TRL)	TRL-7
	Brief Write-up on the Technology Developed	RIGE-SENSE is an indigenous machine-vision–based system developed by C-DAC Kolkata to ensure the quality of rice supplied through Public Distribution Agencies (PDA). Conventional rice age determination using the Mixed Indicator Method is subjective, inconsistent, labor-intensive, and dependent on laboratory-based procedures, rendering it unsuitable for real-time monitoring. To address these limitations, RIGE-SENSE integrates computer vision, AI/ML algorithms, and automated chemical analysis, delivering a rapid, accurate, and objective framework for rice age estimation.
	Details of Contact Person along with Mobile No. and e-mail ID	Gopinath Bej, C-DAC Kolkata 7003776417 gopinath.bej@cdac.in
4.	**Brief details of Technology Developed**	GO-P (*Go-Paryavekshak),* BLE Supported Low Power Consuming Collar Mounted Cattle Health Monitoring Solution.
	Name of the Project under which technology is developed	National Programme on Electronics and ICT Applications in Agriculture and Environment (AgriEnIcs)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-7

Sr. No.	Parameters	Brief details
	Brief Write-up on the Technology Developed	The *Go-Paryavekshak* or *Go-P*, cattle monitoring system, is designed for real-time health and activity monitoring of cows. By attaching a node equipped with an IMU (Inertial Measurement Unit) sensor and a battery to each cow, the system collects data on the cow's movements and behaviour.The IMU sensor captures the x, y, and z-axis movements of each cow and gyroscopic data, which are then analyzed to determine specific activities such as feeding, rumination, lying, and other behaviours. The daily ruminating and feeding time is a significant signature of cattle welfare. It is well acknowledged that eating and rumination time variations can assist farmers in forecasting calving times. Feeding, rumination, and other activity patterns are recognized through machine learning algorithms and monitored the time of these primary parameters. By analyzing the feeding, rumination, and other activity times, different health statuses like healthy, unhealthy, and heat are identified, as these times are significantly different for each category. An automated notification as an SMS has been sent to the registered farm manager after detecting the health conditions i.e. either in heat or illness, enabling prompt intervention. This practical application ensures that farmers can closely monitor their herd's health and reproductive status without needing constant manual observation, thus enhancing farm management efficiency and productivity.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr Hena Ray, C-DAC Kolkata 9830764590, hena.ray@cdac.in
5.	**Brief details of Technology Developed**	GrainEx, an Artificial Intelligence (AI) based Expert Solution for Quality Analysis of Multiple Crops
	Name of the Project under which technology is developed	National Programme on Electronics and ICT Applications in Agriculture and Environment (AgriEnIcs)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-7
	Brief Write-up on the Technology Developed	GrainEx is an AI-based conveyorized machine vision system designed for objective, appearance-based quality analysis of multiple crops, including Tur, Moong, Masoor, Bengal gram, Sorghum, and Wheat. Using a single hardware platform with crop-specific software, it captures high-resolution images of grain samples under controlled conditions via an overhead scanner. The system automatically controls conveyor movement using proximity sensors, segments samples to avoid redundancy, and analyzes quality parameters such as good grains, immature grains, damaged grains, and foreign matter. Results are generated as percentage equivalent weight and presented through a user-friendly GUI with real-time and historical reports.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Amitava Akuli, C-DAC Kolkata +91 9830560405, amitava.akuli@cdac.in

Sr. No.	Parameters	Brief details
6.	**Brief details of Technology Developed**	CT-VIEU, a conveyorized system for assaying of dry red chilli depending upon the appearance-based chilli quality parameters in a rapid and reliable manner.
	Name of the Project under which technology is developed	National Programme on Electronics and ICT Applications in Agriculture and Environment (AgriEnIcs)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-7
	Brief Write-up on the Technology Developed	CT-VIEU is a conveyorised machine vision inspection system developed for quality analysis of dry chilli samples. The system integrates a non-reflective, textureless conveyor belt, a sample feeding and spreading unit, and a machine vision module connected to a PC running Visual C++ based software. Chilli samples are evenly separated using a vibratory discharge chute with optimized metal separators to avoid overlap. High-resolution images are captured under controlled illumination inside an enclosed cabinet, eliminating ambient light interference. The software analyzes the images in real time and displays quality assessment results to the user through an intuitive interface.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Amitava Akuli, C-DAC Kolkata +91 9830560405, amitava.akuli@cdac.in
7.	**Brief details of Technology Developed**	Smart Unified Framework for Automated Quality Assaying & Sorting System for Horticultural Produces (SUFAL), a pendulum bucket type conveyor system for sorting of fruits & vegetables based on colour & size.
	Name of the Project under which technology is developed	National Programme on Electronics and ICT Applications in Agriculture and Environment (AgriEnIcs)
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	TRL-6
	Brief Write-up on the Technology Developed	AI-based image processing technology for 360° surface analysis. Automated sorting by color, size, and defects. Multichannel conveyor systems for seamless handling. Real-time operation for fast and accurate sorting.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Subrata Sarkar, +91 9433111038, subrata.sarkar@cdac.in Dr. Amitava Akuli, +91 9830560405, amitava.akuli@cdac.in C-DAC, Kolkata